

Article

Apatite Geochemistry of the Slyudyanka Deposit, Siberia: Trace Element Composition, Y/Ho Anomaly, and Multivariate Statistical Analysis for Genetic Classification

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Abstract

Apatite is a key indicator mineral whose chemical signature can reveal the genesis and evolution of ore-forming systems. However, correctly interpreting these signatures requires a robust discrimination between apatite types formed by different geological processes, such as metamorphism and hydrothermal activity. This study aims to chemically characterize and genetically classify apatite samples from the Slyudyanka deposit (Siberia, Russia) to establish discriminative geochemical fingerprints for metamorphic and hydrothermal apatite types. We analyzed 80 samples of apatite using total reflection X-ray fluorescence (TXRF) and inductively coupled plasma mass spectrometry (ICP-MS). The geochemical data were processed using principal component analysis (PCA) and k-means cluster analysis to objectively discriminate the apatite types. Our analysis reveals three distinct geochemical groups. Metamorphic veinlet apatite is defined by high U and Pb, low REE, Sr, and Th, and suprachondritic Y/Ho ratios. Massive metamorphic apatite from silicate–carbonate rocks shows extreme REE enrichment and chondritic Y/Ho ratios. Hydrothermal–metasomatic apatite features high Sr, Th, and As, with intermediate REE concentrations and chondritic Y/Ho ratios. Furthermore, we validated the critical and anomalous Y concentrations in the metamorphic veinlet apatite by cross-referencing TXRF and ICP-MS data, confirming the reliability of our measurements for this monoisotopic element. We successfully established diagnostic geochemical fingerprints that distinguish apatite formed in different geological environments at Slyudyanka. The anomalous Y/Ho ratio in metamorphic veinlet apatite serves as a key discriminant and provides insight into specific fractionation processes that occurred during the formation of phosphorites in oceanic environments, which later transformed to apatites during high-grade metamorphism without a change in the Y/Ho ratio. This work underscores the importance of multi-method analytical validation for accurate geochemical classification.

Keywords: TXRF; ICP-MS; apatite; Slyudyanka; PCA; cluster analysis; provenance



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1. Introduction

Apatite is a common accessory mineral in most igneous and metamorphic rocks. The capacity of apatite to incorporate various rare earth and volatile elements makes it a valuable indicator mineral for ore deposit exploration. Furthermore, it is increasingly used as a tool for diagnosing substance sources in sedimentary basins and establishing localization conditions, which is essential for paleogeodynamic reconstructions [1–3]. The

trace element composition of apatite, particularly its rare earth element (REE) content, is of dual significance. It records the mineral's formation history [4–7] while also indicating its potential as a source for these critically important metals, which are vital for the global transition to renewable energy and advanced technologies. The REE composition of apatite has long been utilized as a proxy for ancient seawater chemistry and paleomarine environmental reconstruction, with the expectation that hydrogenous (seawater-derived) REE signals will be preserved [8].

The yttrium-to-holmium (Y/Ho) ratio is a sensitive geochemical discriminant. In igneous systems and related hydrothermal fluids, Y and Ho behave coherently due to their similar ionic radii and charge (CHARAC behavior), maintaining a near-chondritic ratio (~28) [9–13]. In contrast, oxidative scavenging in seawater fractionates this pair, producing characteristic super-chondritic Y/Ho ratios (>52) [14]. Studies of hydrothermal and magmatic apatite confirm its diagnostic utility; for example, Zhuxi hydrothermal apatite has suprachondritic Y/Ho, while associated magmatic apatite is chondritic [15–17]. Non-chondritic Y/Ho ratios and tetrad effects in apatite can signal advanced magmatic fractionation, fluid interaction, and rare metal mineralization potential [9,11,15,17–20].

However, the systematic geochemical characterization of apatite formed in high-grade metamorphic terrains, and its discrimination from apatite of clear hydrothermal affiliation within the same complex, is less well-established. The Slyudyanka crystalline complex (Siberia, Russia) presents an ideal natural laboratory to address this problem. It is a granulite facies terrane containing both metamorphic apatite (derived from sedimentary phosphorites and silicate–carbonate rocks) and later hydrothermal–metasomatic apatite in phlogopite veins [21]. This study aims to (1) establish diagnostic geochemical fingerprints for the three main apatite types (hydrothermal–metasomatic, metamorphic veinlet, massive metamorphic) in the Slyudyanka deposit and (2) evaluate the persistence of the Y/Ho anomaly as a tracer for a marine sedimentary protolith through granulite facies metamorphism.

For the analysis of apatite, analytical methods are essential. One of the perspective methods for apatite analysis is total reflection X-ray fluorescence (TXRF), a microanalytical technique requiring minimal sample mass and no calibration standards [22–25]. Our prior work has optimized TXRF for multi-element analysis of apatite across various size ranges via a direct acid digestion preparation. The method demonstrates accuracy comparable to inductively coupled plasma mass spectrometry (ICP-MS) with sufficient sensitivity for geochemical studies, offering a rapid and cost-effective analysis. Thus, prior to proceeding to geochemical interpretations, the third aim was to demonstrate the efficacy of TXRF as a primary, cost-effective analytical tool for such provenance studies, validated against ICP-MS. By resolving these aims, we provide a new geochemical framework for apatite classification in complex metamorphic settings and contribute to understanding fluid–rock interaction and sedimentary heritage in the Central Asian Orogenic Belt (CAOB).

2. Geological Setting, Materials, and Methods

2.1. Geological Setting

The Slyudyanka crystalline complex includes the most highly metamorphosed (granulite facies) part of the Khamar-Daban terrane of the CAOB, directly adjacent to the southern Siberian Craton [21] (Figure 1A). The Slyudyanka complex is a type locality of the Early Paleozoic Baikal Collisional Belt [26]. Geological maps of the two studied areas of the Slyudyanka Complex are shown in Figure 1B,C.

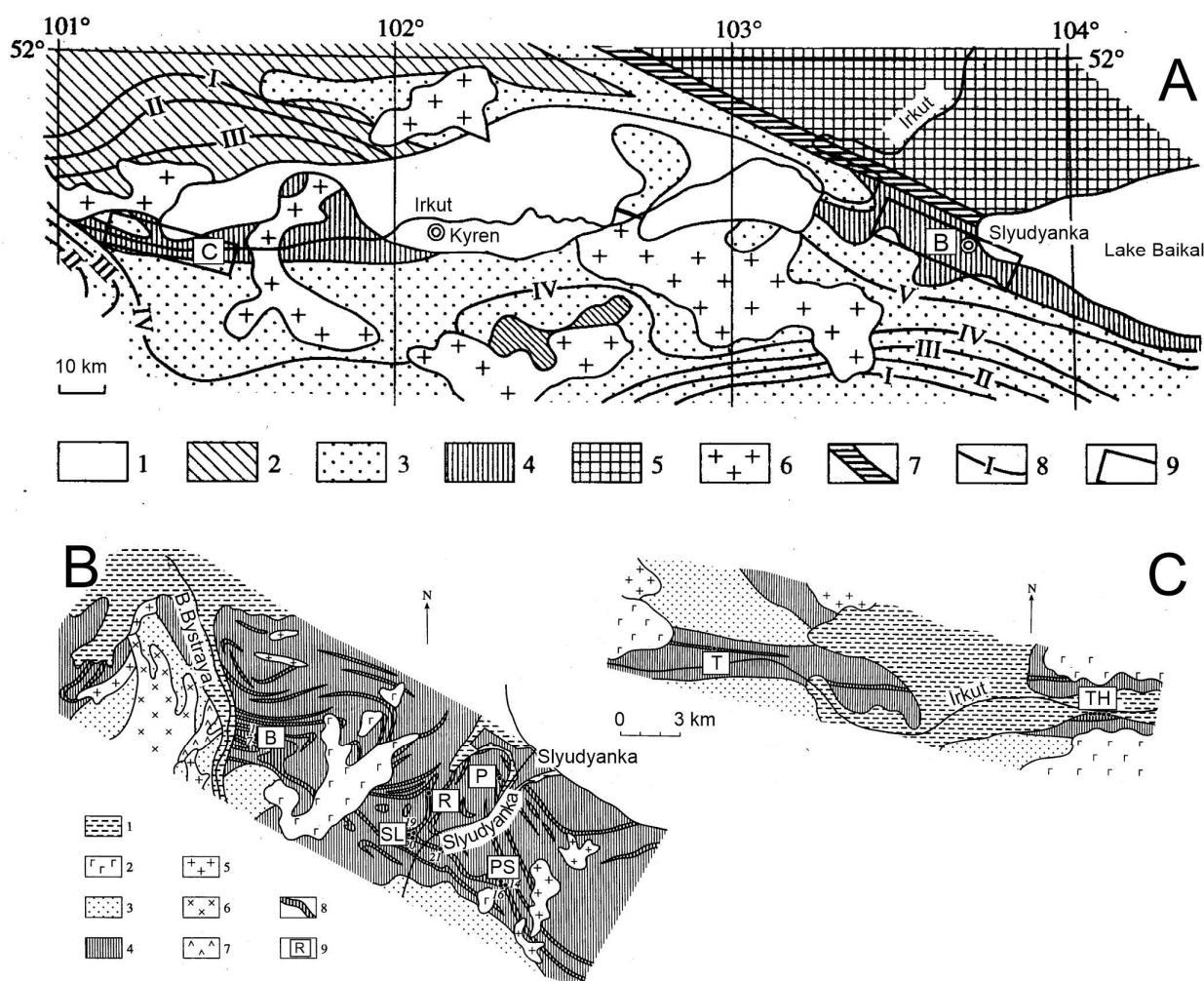


Figure 1. Tectonic structure of the Slyudyanka metamorphic complex (A) with enlarged geological maps of regions within the Slyudyanka settlement (B) and Obrub-Haradaban area (C) (compiled based on [27]). Symbols in (A): 1—Cenozoic sediments; 2—Ediacaran–early Paleozoic sediments; 3–4—metamorphic complexes of Hangarul and Slyudyanka series, respectively; 5—Archean–early Proterozoic metamorphic rocks of the Siberian Craton; 6—Paleozoic granites; 7—Major Sayan Fault; 8—metamorphic mineral zones (I—garnet, II—staurolite, III—sillimanite, IV—K-feldspar, V—hypersthene); 9—enlarged geological maps. Symbols in (B,C): 1—Cenozoic sediments; 2—Cenozoic basalts; 3–4—metamorphic complexes of Hangarul and Slyudyanka series, respectively; 5–7—early Paleozoic granites, syenites, and gabbros, respectively; 8—horizons of metaphosphates; 9—sampling sites with indexes as in Table 1.

The Slyudyanka complex is located within the CAO, near the southern end of Lake Baikal. The peak of metamorphism at Slyudyanka was estimated in a time range of ~495–477 Ma, depending on isotope dilution thermal ionization mass spectrometry (ID-TIMS) or laser ablation (LA) ICP-MS dating techniques and using igneous or metamorphic rocks [28–30]. After the granulite peak of metamorphism, the Slyudyanka complex underwent a complex thermal history with the youngest K–Ar closure age for phlogopite at ~270 Ma [31]. The most detailed description of the layered strata, structure, magmatism, metamorphism, and evolution of the Slyudyanka complex is provided in a monograph [21]. The 80 apatite samples analyzed in this study are legacy specimens collected from the Slyudyanka complex during geological fieldwork in the early 1980s. While precise geographic coordinates are not available, their genetic context is well-constrained by original sample labels, associated mineral assemblages, and the extensively mapped geology of the region [27]. Two main groups were distinguished from the selected apatite samples: meta-

morphic and metasomatic. The metamorphic group consists of two subgroups. The first subgroup includes apatite crystallized from metasedimentary rocks (metaphosphorites). Phosphate-bearing siliceous dolomites underwent metamorphism under granulite facies conditions at a temperature of ~800 °C and a pressure of ~6 kbar. These are sheet bodies that occur as a part of the Slyudyanka Series. The rocks are usually banded; bands (layers) of quartz, diopside, and apatite alternate in composition. Samples of the metamorphic type are designated by the indices «SL», «TH», «T», and «B». The second subgroup, called the silicate–carbonate (calciphyre) type, consists of nests or inclusions of apatite in sheet metamorphic rocks composed of green diopside. These samples are labeled as «B-5», fine-crystalline apatite from a nest of green rocks, and «PS-4», black fine-crystalline varieties. The metasomatic group includes apatite from phlogopite-bearing veins, which provides the name of the whole complex (Slyudyanka means micaceous in Russian). These are usually large-crystalline varieties ranging in size from several cm to several dm. They vary in color—blue, greenish, and sometimes with a violet tint. They are labeled as «PS», «P», and «R». Figure 2 shows pictures of three different types of apatite: hydrothermal–metasomatic, metamorphic apatite, and massive metamorphic apatite of silicate–carbonate rocks.

Table 1. Concentrations (µg/g) of Slyudyanka apatite samples determined by the TXRF method. Samples with code names B, P, PS, SL, and R are from the Kultuk formation, and T and TH are from the Perevalnaya formation.

Sample	Genetic Type	Mn	As	Sr	Y	La	Ce	Pb	Th	U
B-2307-A	Metamorphic	8.4	10.4	580	24.0	<LOD	21.7	11.0	14.6	154
B-233-A	Metamorphic	39.3	3.2	601	23.1	<LOD	33.0	26.4	17.6	107
B-2457-A	Metamorphic	13.9	6.4	689	43.6	<LOD	19.4	33.6	3.1	252
B-592-A	Silicate–carbonate	244	<LOD	191	1702	156	873	6.5	<LOD	15.5
B-595-A2	Silicate–carbonate	271	1.4	356	774	272	817	8.1	<LOD	14.7
B-513-A5	Silicate–carbonate	182	<LOD	436	882	621	2020	3.9	10.6	0.0
P-701-A	Metasomatic	49.4	6.2	1490	32.1	<LOD	67.2	23.6	85.7	66.3
P-706-A	Metasomatic	106	16.1	1506	41.7	<LOD	53.5	10.3	98.9	77.4
P-708-AG	Metasomatic	48.8	6.3	1484	32.0	<LOD	60.8	23.9	90.1	64.8
P-709-A	Metasomatic	42.1	8.7	1421	27.5	<LOD	56.6	5.3	85.2	61.1
P-712-A	Metasomatic	41.9	12.0	1463	23.4	<LOD	60.9	9.9	82.9	66.9
P-714	Metasomatic	36.8	14.0	1271	35.0	76.0	89.0	20.0	124	61.0
P-720-A	Metasomatic	44.1	13.5	1403	47.6	<LOD	83.7	3.4	67.9	59.0
P-720-A1	Metasomatic	43.0	16.6	1495	53.7	19.3	79.5	5.5	72.4	57.1
P-726-AB	Metasomatic	55.3	6.4	1140	50.3	<LOD	70.7	13.8	66.4	40.0
P-726-AB1	Metasomatic	57.5	9.7	1257	49.6	<LOD	52.7	3.4	75.4	40.5
P-726-AG	Metasomatic	83.4	9.8	1222	43.0	<LOD	51.3	12.2	69.8	43.9
P-747	Metasomatic	51.2	12.0	1352	38.0	66.0	66.0	20.0	100	55.0
P-789-AC	Metasomatic	43.6	35.7	1371	48.6	43.8	130	11.2	72.8	110
P-798	Metasomatic	60.7	26.0	1404	56.0	80.0	118	23.0	111	64.0
P-804-A	Metasomatic	72.7	15.4	1228	38.1	30.4	58.2	3.8	111	61.4
P-804-A	Metasomatic	73.2	15.4	1217	37.8	<LOD	68.9	5.9	92.3	56.9
P-810	Metasomatic	62	18.0	1358	46.0	71.0	122	24.0	124	75.0
P-811-AP	Metasomatic	64.5	14.1	1428	54.0	<LOD	83.1	5.2	111	62.0
P-811-AC	Metasomatic	80.4	22.4	1398	58.3	<LOD	106	7.3	104	64.3
P-832-AZ	Metasomatic	43.0	20.9	1354	51.3	<LOD	102	15.0	95.8	69.5
P-832-A3	Metasomatic	42.6	11.8	1328	53.9	<LOD	99.6	19.3	93.7	73.4
P-833-AC	Metasomatic	38.4	7.3	1320	42.4	24.6	89.2	7.0	85.2	38.5
P-853-A31	Metasomatic	33.3	13.6	1646	66.1	37.5	142	7.1	77.4	65.0
P-853-A32	Metasomatic	35.1	16.1	1693	64.6	89.1	176	2.5	70.2	44.9
P-853-AF	Metasomatic	50.7	10.6	1449	75.3	<LOD	129	17.8	80.9	60.9
P-853-AG	Metasomatic	71.9	11.4	1346	46.9	20.4	80.7	17.2	74.4	57.4
P-853-AΓ	Metasomatic	74.5	14.2	1280	45.7	29.0	89.1	4.1	70.8	56.3
P-853-AC	Metasomatic	40.5	9.5	1502	45.6	<LOD	92.9	8.4	96.3	59.4
P-864	Metasomatic	85	11.0	1295	26.0	46.0	63.0	19.0	90.0	56.0
P-893-A	Metasomatic	65.4	19.6	1315	45.9	<LOD	79.8	18.3	121	54.1

Table 1. Cont.

Sample	Genetic Type	Mn	As	Sr	Y	La	Ce	Pb	Th	U
P-923-A	Metasomatic	46	30.0	1602	104	267	494	32.0	151	133
P-934-A	Metasomatic	60.6	13.8	1410	58.4	<LOD	46.5	5.1	86.0	52.9
PS-312-AG	Metasomatic	20.0	9.0	1892	46.0	220	319	29.0	131	56.0
PS-313	Metasomatic	13.5	9.5	2158	55.6	127.4	306	29.0	120	64.2
PS-313-AC	Metasomatic	14.5	11.9	1803	47.4	92.2	271	11.3	103	37.5
PS-314-AG	Metasomatic	17.9	13.0	1778	34.0	129	177	20.0	118	49.0
PS-316	Metasomatic	10.0	8.7	1895	58.6	90.7	317	25.0	108	58.5
PS-317	Metasomatic	11.8	12.1	1530	42.9	118	236	8.1	87.1	45.0
PS-317-AC	Metasomatic	14.9	11.0	1990	38.9	148	287	4.4	118	49.2
PS-318	Metasomatic	12.7	13.1	1823	36.8	46.2	194	13.4	87.8	51.4
PS-318-AG	Metasomatic	16.9	8.8	1928	31.0	46.7	168	15.0	86.5	29.2
PS-319-A	Metasomatic	21.0	14.0	1466	25.0	84.0	96.0	18.0	96.0	44.0
PS-334-A	Metamorphic	201	7.6	364	4.7	<LOD	35.6	11.9	<LOD	82.6
PS-353	Metamorphic	5.0	4.0	557	64.0	34.0	49.0	46.0	10.0	220
PS-354-A	Metamorphic	<LOD	5.6	1094	51.7	<LOD	<LOD	28.9	<LOD	208
PS-355-A	Metamorphic	195	4.5	326	4.6	<LOD	29.0	5.5	3.0	79.0
PS-406-AMG	Silicate–carbonate	653	<LOD	1202	466	1017	1942	2.2	8.8	9.6
PS-414-AMG	Silicate–carbonate	524	4.7	1578	705	691	2048	3.2	31.0	16.8
PS-573-A	Metasomatic	68	12.0	2028	38.0	105	134	26.0	125	103
R-1-A	Metasomatic	38.9	15.6	1252	50.5	23.0	98.3	<LOD	107	50.7
R-712-A1	Metasomatic	48.3	7.7	1484	25.2	<LOD	61.0	22.0	88.2	58.7
SL-254-A	Metamorphic	23.0	2.5	500	29.7	50.2	52.8	30.3	<LOD	116
SL-257-A	Metamorphic	55.5	8.6	343	51.9	<LOD	<LOD	10.7	3.7	123
SL-258-A	Metamorphic	12.8	9.8	499	26.4	<LOD	21.2	9.6	<LOD	125
SL-264	Metamorphic	13	<LOD	507	43.0	22.0	22.0	32.0	7.0	218
SL-265-A	Metamorphic	25.2	<LOD	733	19.6	<LOD	33.0	19.3	<LOD	223
SL-266-A	Metamorphic	64.0	4.8	287	2.7	<LOD	<LOD	11.2	<LOD	50.4
SL-271-A	Metamorphic	4.9	4.8	461	25.7	<LOD	<LOD	4.8	<LOD	82.0
SL-271-A5	Metamorphic	<LOD	4.8	450	19.5	<LOD	20.2	9.6	5.9	84.3
SL-273-A1	Metamorphic	18.3	3.4	577	15.7	<LOD	26.3	16.7	<LOD	154
SL-276	Metamorphic	<LOD	<LOD	712	67.0	54.0	45.0	38.0	63.0	119
SL-288-AG	Metamorphic	29.2	<LOD	518	26.5	<LOD	20.4	22.4	<LOD	98
SL-291-AG	Metamorphic	17.8	7.8	541	60.9	<LOD	21.8	23.3	<LOD	300
SL-291-A3	Metamorphic	16.7	11.4	578	66.4	<LOD	24.1	19.6	<LOD	320
T-93-A	Metamorphic	31.3	2.9	388	10.8	<LOD	<LOD	9.0	<LOD	52.9
T-93-A2	Metamorphic	25.8	<LOD	374	8.3	<LOD	<LOD	10.4	<LOD	52.3
T-95-A	Metamorphic	46.9	<LOD	440	10.3	<LOD	<LOD	15.4	<LOD	33.9
T-95-A1	Metamorphic	17.6	0.0	508	2.5	<LOD	<LOD	9.1	<LOD	32.5
T-97-A	Metamorphic	38.0	3.9	470	8.7	<LOD	<LOD	13.5	<LOD	38.9
T-97-A1	Metamorphic	40.9	0.0	494	10.0	<LOD	33.3	4.4	<LOD	35.5
TH-701-A	Metasomatic	44.9	5.3	1530	38.9	22.1	77.7	27.9	88.6	70.3
TH-715	Metamorphic	34	<LOD	321	32.0	14.0	<LOD	18.0	8.0	55.0
TH-74-A	Metamorphic	21.6	3.8	429	39.0	<LOD	<LOD	18.3	<LOD	69.8
TH-78-A1	Metamorphic	22.3	2.9	335	5.4	<LOD	21.7	10.1	<LOD	30.3

2.2. Analytical Instrumentation

2.2.1. Total Reflection X-Ray Fluorescence (TXRF) Analysis

Elemental analysis via TXRF was conducted using a benchtop S2 PICOFOX spectrometer (Bruker Nano Analytics, Berlin, Germany). The instrument is equipped with a Mo-anode X-ray tube, a multilayer monochromator for beam conditioning, and a 30 mm² silicon drift detector with an energy resolution of <150 eV for the Mn K α line. Measurement parameters were standardized. The live time for one analysis was 500 s, with the X-ray tube operating at 50 kV and 0.50 mA. Samples were prepared on quartz carriers, which served as both sample holders and reflectors. The acquired spectra were processed using the vendor's «Spectra 7.8.2» software.

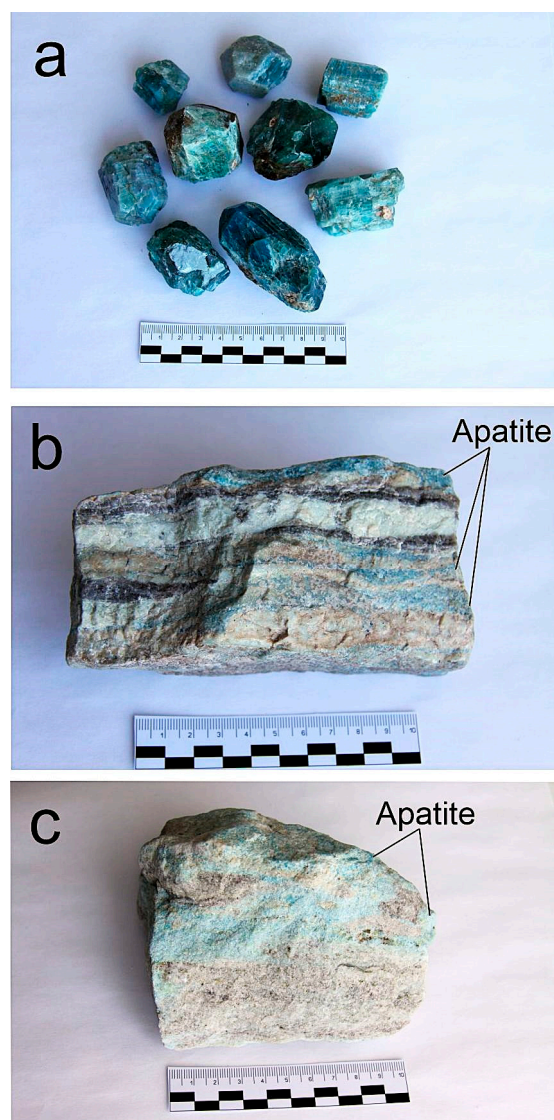


Figure 2. Slyudyanka apatite: (a) hydrothermal–metasomatic apatite crystals of various colors (dark, green, colorless) of phlogopite veins, (b) veinlets of metamorphic apatite of deep blue color crystallized from metaphosphorites, (c) massive metamorphic apatite of light blue color from silicate–carbonate rocks.

2.2.2. Inductively Coupled Plasma Mass Spectrometry (ICP-MS) Analysis

A subset of representative samples was analyzed for their rare earth element (REE) content using an Agilent 7900 quadrupole ICP-MS (Agilent Technologies Inc., Santa Clara, CA, USA). The instrument was operated in standard mode. Sample introduction was achieved using a MicroMist nebulizer (Glass Expansion, Melbourne, Australia) coupled with a Scott-type spray chamber (Glass Expansion, Melbourne, Australia) maintained at a constant temperature by a Peltier cooling element (Glass Expansion, Melbourne, Australia). Standard nickel sampler and skimmer cones were used for the interface.

2.3. Reagents

All acids—hydrofluoric acid (HF, 48%), nitric acid (HNO₃, 65%), and perchloric acid (HClO₄, 70%–72%)—were of ultrapure grade (Merck, Darmstadt, Germany). Single-element stock solutions of gallium (Ga), indium (In), and bismuth (Bi) (each at 1000 mg/L, Merck, Darmstadt, Germany) were used for internal standardization. All dilutions were performed using ultrapure deionized water (18.2 MΩ·cm, Elga Labwater, High Wycombe, UK).

2.4. Sample Preparation

2.4.1. TXRF

Apatite samples were first mechanically crushed and homogenized using a mortar grinder (Retsch RM-200, RETSCH GmbH, Haan, Germany). For digestion, approximately 10 mg of the resulting powder was transferred to a PTFE vessel. To this, 100 µL of concentrated HNO₃ was added, and the sealed vessel was heated on an electric hotplate at 160 °C for 30 min. After cooling, the solution was diluted with 800 µL of ultrapure water and spiked with 100 µL of a Ga internal standard solution. A precise aliquot of this final solution was then deposited onto a quartz carrier and dried on a heating plate to form a thin film for analysis.

2.4.2. ICP-MS

For ICP-MS analysis, a larger aliquot of 50 mg of powdered apatite was digested in a 35 mL PTFE beaker using a mixture of 5 mL HF, 2 mL HNO₃, and 1 mL HClO₄. The beakers were swirled for homogenization and left at room temperature for 24 h to ensure the initial reaction. A stepped heating program was then applied: 3 h at 110 °C, followed by 3 h at 140 °C, and a final 3 h at 180 °C. After evaporation to incipient dryness, the residues were re-treated with 1 mL of water and 1 mL of HNO₃ and evaporated again. The final residue was dissolved in 10 mL of 10% (v/v) HNO₃ with gentle heating to yield a clear solution. This solution was transferred to a 50 mL volumetric flask and made up to volume with ultrapure water, resulting in a primary stock solution. A subsequent dilution was performed by taking a 1 mL aliquot of this stock, diluting it with 4 mL of 2% (v/v) HNO₃ in a polystyrene tube, yielding a total dilution factor of 5000. Immediately before analysis, internal standards (In and Bi) were added to all solutions at a final concentration of 10 ng/mL [32].

2.5. Quantification and Data Treatment

2.5.1. TXRF Quantification and Figures of Merit

Elemental concentrations (C_i , in mg/kg) were calculated using the following established equation:

$$C_i = \frac{C_{is} \cdot N_i \cdot S_{is}}{N_{is} \cdot S_i} \quad (1)$$

where C_{is} is the internal standard concentration, N_i and N_{is} are the net peak areas of the analyte and internal standard, and S_i and S_{is} are their respective relative sensitivities.

The limit of detection (LOD) for each element was derived from the spectrum using the following formula:

$$C_{LOD} = 3C_i \sqrt{N_{BG}} / N_i \quad (2)$$

where N_{BG} is the background count under the analyte's peak.

The elements were chosen according to the sensitivity restriction of TXRF because some REEs were not determined. The following elements were determined by TXRF: Mn, As, Sr, Y, La, Ce, Pb, Th, and U. The detection limits for the TXRF method ranged from 1 µg/g for Mn, As, Sr, Y, Pb, Th, and U; for La and Ce, they were 15 µg/g. The measurement combined expanded uncertainties (U_{CA}) for Y; Th did not exceed 3.9%, Sr and U did not exceed 6.9%, Mn, As, La, and Ce did not exceed 9.9%, and Pb did not exceed 16.6%. The analytical figures of merit were calculated for the TXRF method in our previous publication [22].

2.5.2. ICP-MS Performance

The uncertainty of the ICP-MS measurements was maintained below, 5% for Y, 7% for light REEs, and 10% for heavy REEs, following established laboratory procedures [32].

Method detection limits ranged from 0.021 to 0.24 µg/g for middle and heavy REEs and from 0.44 to 2.0 µg/g for light REEs.

2.6. Statistical Analysis

All multivariate statistical analyses, including principal component analysis (PCA) and k-means clustering, were performed using Origin 2024 software (OriginLab Corporation, Northampton, MA, USA). Prior to multivariate analysis, concentration data below the LOD were replaced with a value of $\text{LOD}/\sqrt{2}$, a common practice to minimize bias in statistical treatment of censored data. The dataset was then auto-scaled (mean-centered and divided by the standard deviation for each variable) to ensure all elements contributed equally to the PCA and cluster analysis, regardless of their absolute concentration ranges.

3. Results

3.1. TXRF and ICP-MS Results

Table 1 presents the concentrations of 80 apatite samples from Slyudyanka obtained using the TXRF method. Figure 3 shows the spectra of apatite samples that differ most in elemental composition. For the correct interpretation of spectra, all intensities were normalized to the $\text{GaK}\alpha$ line of the internal standard Ga, which was added in the same quantity to all samples. From Figure 3, we can observe a large variety of intensities obtained for different types of apatite. The full range of elements identified in the samples, from P to U, is displayed in the spectra. The air between the sample and the detector causes the Ar peak to appear, the addition of the internal standard gives the Ga peak, and the Mo-anode X-ray tube's scattering radiation causes the Mo peak. The spectra also showed a significant amount of Si, which can be explained by the use of quartz carriers as sample holders and reflectors.

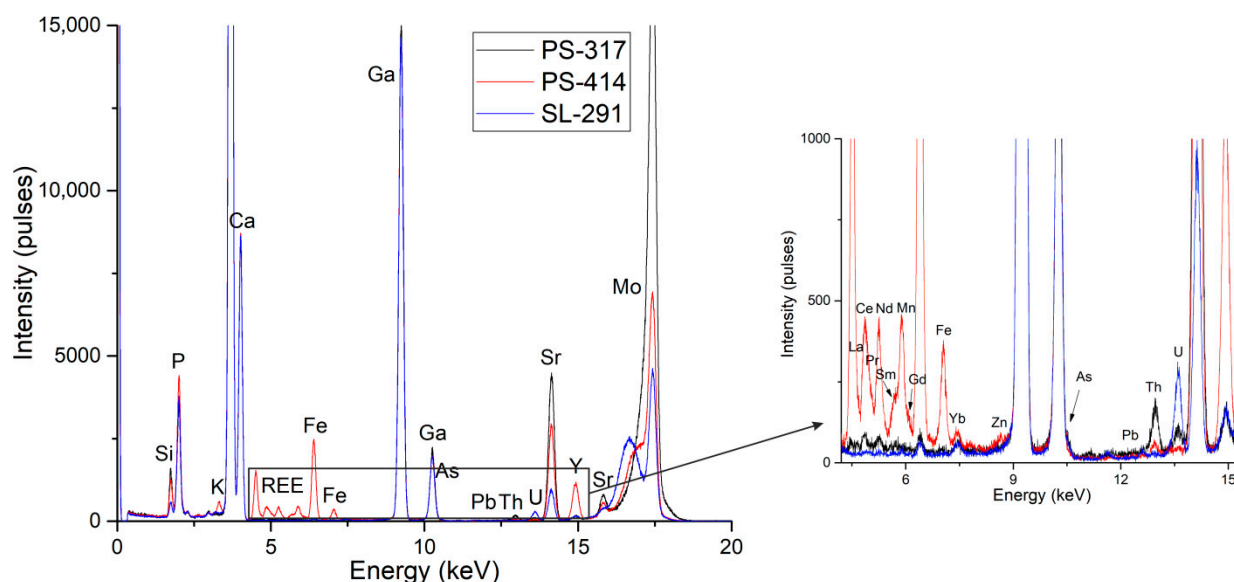


Figure 3. TXRF spectra, normalized to $\text{GaK}\alpha$, of apatite samples: PS-317—hydrothermal-metasomatic apatite of phlogopite veins; PS-414—massive metamorphic apatite of silicate-carbonate rocks; SL-291—metamorphic veinlet apatite of metaphosphorites.

The results of 80 samples were observed, and the representative parts, including 17 samples, were measured using ICP-MS. The samples were chosen according to the classification of apatite types based on TXRF data. Table 2 presents the REE + Y data. ICP-MS was applied to determine the rare earth element (REE) composition for subsequent

geochemical research. A comparison of the data obtained using the two methods showed good agreement between the results.

Table 2. Concentrations ($\mu\text{g/g}$) of REEs in Slyudyanka apatite samples determined by the ICP-MS method. Samples with code names B, P, PS, SL, and R are from the Kultuk formation, and T and TH are from the Perevalnaya formation.

Sample	Genetic Type	Y	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
B-513-A5	Sil.–carbonate	661	693	1584	216	886	264	18.9	227	32.6	163	28.2	64.9	7.9	40.4	5.3
P-708-AG	Metasomatic	32.4	59.5	105	10.4	35.0	6.9	1.3	6.1	0.9	5.2	1.0	2.6	0.3	2.0	0.3
P-714	Metasomatic	33.8	62.5	110	11.2	39.3	7.4	1.6	6.6	1.0	5.8	1.2	3.2	0.4	2.6	0.3
P-720-A1	Metasomatic	50.3	76.2	134	14.1	47.2	9.0	2.0	8.4	1.3	7.7	1.6	4.1	0.6	3.2	0.4
P-798	Metasomatic	56.4	70.7	126	14.8	53.0	9.9	2.2	9.3	1.5	9.1	1.8	4.8	0.6	3.6	0.5
P-832-AZ	Metasomatic	59.0	92.5	169	16.9	56.1	10.5	2.4	9.9	1.6	9.3	1.9	4.9	0.7	4.0	0.5
P-923-A	Metasomatic	91.0	247	421	42.0	141	24.1	4.9	20.6	3.0	16.0	3.0	7.8	1.0	5.6	0.7
PS-312-AG	Metasomatic	42.2	211	316	28.1	85.2	12.7	3.0	10.5	1.5	7.4	1.3	3.4	0.5	2.6	0.3
PS-314-AG	Metasomatic	30.2	114	180	17.0	54.5	8.6	2.0	7.4	1.0	5.4	1.0	2.6	0.3	1.9	0.2
PS-319-A	Metasomatic	23.7	75.9	123	12.0	39.3	6.7	1.5	5.8	0.8	4.2	0.8	2.1	0.3	1.6	0.2
PS-354-A	Metamorphic	42.2	20.2	34.0	3.5	12.2	2.6	0.6	3.0	0.5	3.3	0.7	2.1	0.3	1.9	0.3
PS-406-AMG	Sil.–carbonate	328	839	1335	126	495	101	22.2	86.7	12.2	65.6	12.3	31.4	4.1	25.1	4.2
PS-414-AMG	Sil.–carbonate	528	641	1244	142	543	135	29.3	120	18.3	101	19.9	51.1	6.8	39.0	6.0
PS-573-A	Metasomatic	33.3	97.1	154	14.9	50.1	8.6	1.8	7.6	1.1	5.9	1.1	3.0	0.4	2.3	0.3
SL-254-A	Metamorphic	31.0	12.8	23.0	2.4	8.8	2.0	0.5	2.3	0.4	2.8	0.6	1.8	0.3	1.6	0.2
SL-257-A	Metamorphic	50.5	22.8	43.1	4.6	17.0	3.7	0.9	4.3	0.7	4.8	1.1	3.0	0.4	2.7	0.4
T-97-A	Metamorphic	12.4	9.4	15.7	1.5	5.1	1.1	0.3	1.2	0.2	1.2	0.3	0.7	0.1	0.6	0.1

3.2. Multivariate Analysis

Multivariate analysis is a common method for the discrimination of apatite of different origins [33]. Different statistical methods, including principal component analysis, partial least squares [33], support vector machine [7], and decision tree algorithms [34], are used to classify apatite samples.

We applied a convenient chemometric technique called principal component analysis (PCA). The study of the score and loading scatter plots allowed the identification of the criteria for clustering objects. PCA was applied to the elemental composition obtained for the apatite samples, which enabled us to classify them into three main groups. Figure 4 presents the score scatter plots constructed in projections of the first and second principal components of PC1-PC2. The cumulative percentage of variance in the correlation matrix for the first two principal components was greater than or equal to 67% for all scatter plots. Correlation analysis showed a positive correlation between Th and Sr ($r = 0.88$), La and Ce ($r = 0.95$), and Ce and Y ($r = 0.72$). The following elements, Ce, La, and Y, were the most significant variables for the first PC. The second PC was characterized by the dispersion of Th and Sr.

The samples with relatively high U concentrations were in the metamorphic group, and those with high Sr and Th concentrations were in the hydrothermal–metasomatic group. The sample, which belongs to the latter group, differs in REE content and is assigned to apatite of silicate–carbonate rocks. The PCA results showed that the obtained groups of apatite samples were consistent with the geochemical data on the genesis of these samples.

Application of generalized cluster analysis by k-means is a fast and easy way for provenance analysis without preliminary preparation of the dataset, for example, excluding outliers. This method was applied under the following conditions: the k-means algorithm, the squared Euclidean distance method, and a training/test samples ratio of 2:1. Table 3 presents the results of CA of apatite samples, characterized by type. The total number of clusters is equal to 3, as well as the actual quantity of apatite types.

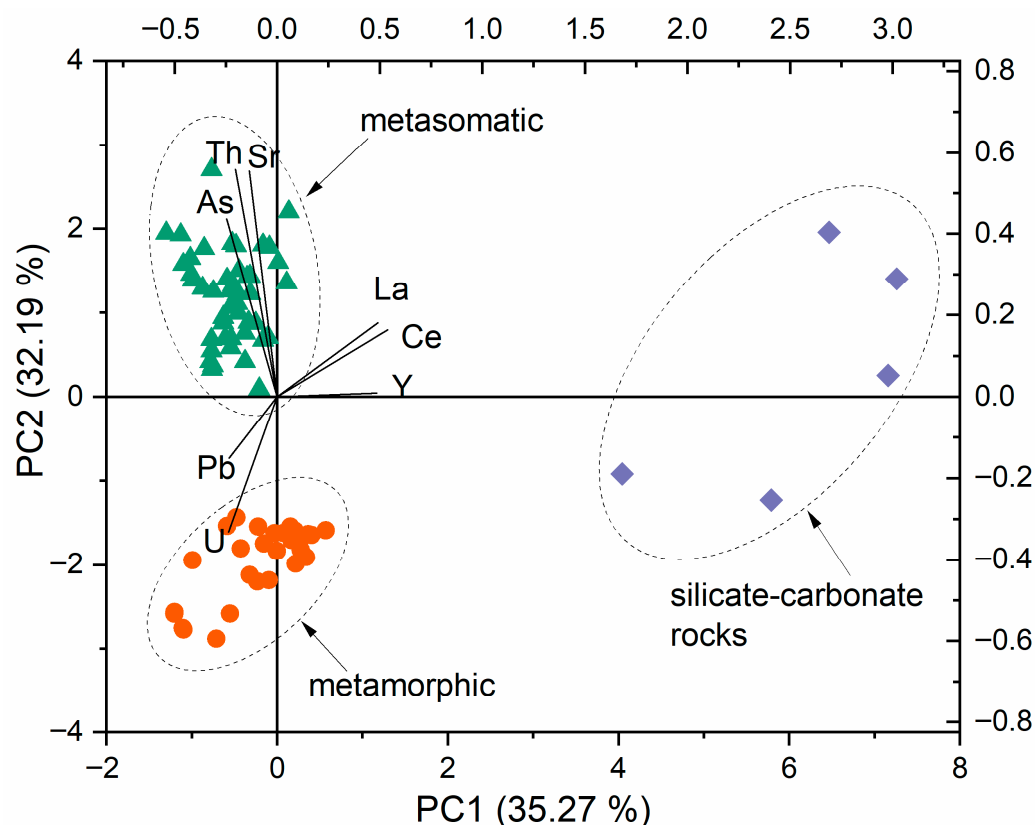


Figure 4. PCA results based on the TXRF element composition of apatite samples from Slyudyanka. Three apatite groups were obtained by their genesis: the metamorphic group is highlighted in red; the hydrothermal–metasomatic group is highlighted in green; and blue rhombs are apatite of silicate–carbonate rocks, whose origin is also metamorphic.

Table 3. Frequency table for cluster analysis of apatite samples characterized by type.

Apatite Type	Cluster Number				Total
	1	2	3		
Hydrothermal–metasomatic apatite	0	47	0		47
Metamorphic apatite veinlets of metaphosphorites	28	0	0		28
Massive metamorphic apatite of silicate–carbonate rocks	0	0	5		5

3.3. Geochemistry of Slyudyanka Apatite

To study the geochemistry of apatite from Slyudyanka, we used data on the rare element composition obtained by ICP-MS (Table 2). Chondrite-normalized and PAAS-normalized diagrams were constructed for Slyudyanka apatite (Figure 5). At first glance, there are significant differences among the three types of apatite.

Figure 5a shows the chondrite-normalized diagram for the three groups of apatite samples. Metasomatic apatite shows a regular trend from high to low REEs according to the atomic number of elements, with a small negative Eu anomaly. Metamorphic apatite is characterized by lower concentrations of light and middle REEs and a more irregular pattern with a small negative Eu anomaly and a notable positive Y anomaly. The values of heavy REEs of metamorphic apatite were comparable to those of metasomatic apatite. Apatite from silicate–carbonate rocks has significantly higher REE content than metasomatic and metamorphic apatite samples. Two analyzed samples of this type have no Eu anomaly, and one studied sample has a very deep negative Eu anomaly.

Figure 5b shows the REE normalization to Post-Archean Australian Shale (PAAS). This normalization reflects the REE depletion of metamorphic apatite relative to an aver-

age marine sedimentary rock represented by PAAS. Metasomatic apatite is only slightly REE-enriched relative to PAAS, whereas apatite of silicate–carbonate rocks is highly REE-enriched at the level of REE–ore concentrations.

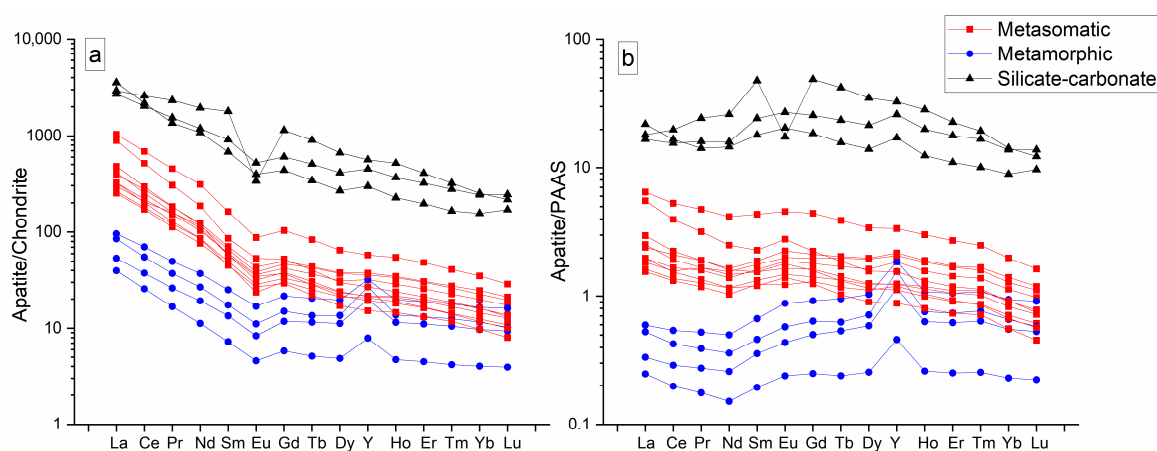


Figure 5. Chondrite-normalized (a) and PAAS-normalized diagrams (b) obtained based on the ICP-MS results of measurements of Slyudyanka apatite.

4. Discussion

The distinct trace element signatures of the three apatite groups are controlled by differing mineral stabilities and fluid compositions during their formation. For the metamorphic veinlet apatite, the high U and low REE, Th, and Sr concentrations are a direct function of high-grade metamorphic conditions. During granulite facies metamorphism, the stability of monazite effectively sequesters the available light REE and Th [35], while plagioclase incorporates Sr [36]. Consequently, apatite crystallizing in this environment is depleted in these elements. Uranium, being less compatible in the structures of monazite and common rock-forming minerals, partitions into the apatite structure. The high Pb content is predominantly radiogenic, accumulating from the in situ decay of U over geological time. In contrast, the hydrothermal–metasomatic apatite derives its composition from an evolved fluid. Its high Sr content suggests interaction with, or derivation from, a carbonate-rich or plagioclase-bearing source. Concurrently, elevated Th and As point to a metasomatic fluid enriched in incompatible and metalloid elements, consistent with its formation in phlogopite-bearing veins. The extreme REE enrichment in the massive metamorphic apatite from silicate–carbonate rocks indicates a protolith and formation environment where competing REE host minerals, like monazite, were absent. In this setting, apatite became the primary sink for all available REEs during metamorphic recrystallization.

The clear separation between the two metamorphic groups on the PCA plot (Figure 4) is a direct reflection of their distinct protoliths. The veinlet apatite (red) inherited a high-U, low-REE signature from a marine phosphorite precursor where REEs were sequestered by monazite during metamorphism, whereas the massive apatite (blue) formed from an REE-rich calc–silicate rock where apatite became the dominant REE host.

The main feature of metamorphic apatite is a positive anomaly of Y relative to Ho. Yttrium and Ho are two elements with similar geochemical affinity [9–12]. Therefore, deviation of the Y/Ho ratio from the chondritic value requires special analytical control. To verify the trueness of the analytical results, Y from the same samples was determined by the ICP-MS and TXRF methods (Figure 6), which showed good correlation over the entire concentration range. The correlation coefficient r was 0.9966, and the measured concentration range was from 12 to 104 $\mu\text{g/g}$. Thus, the anomalously high, suprachondritic Y/Ho ratio is a characteristic of metamorphic apatite.

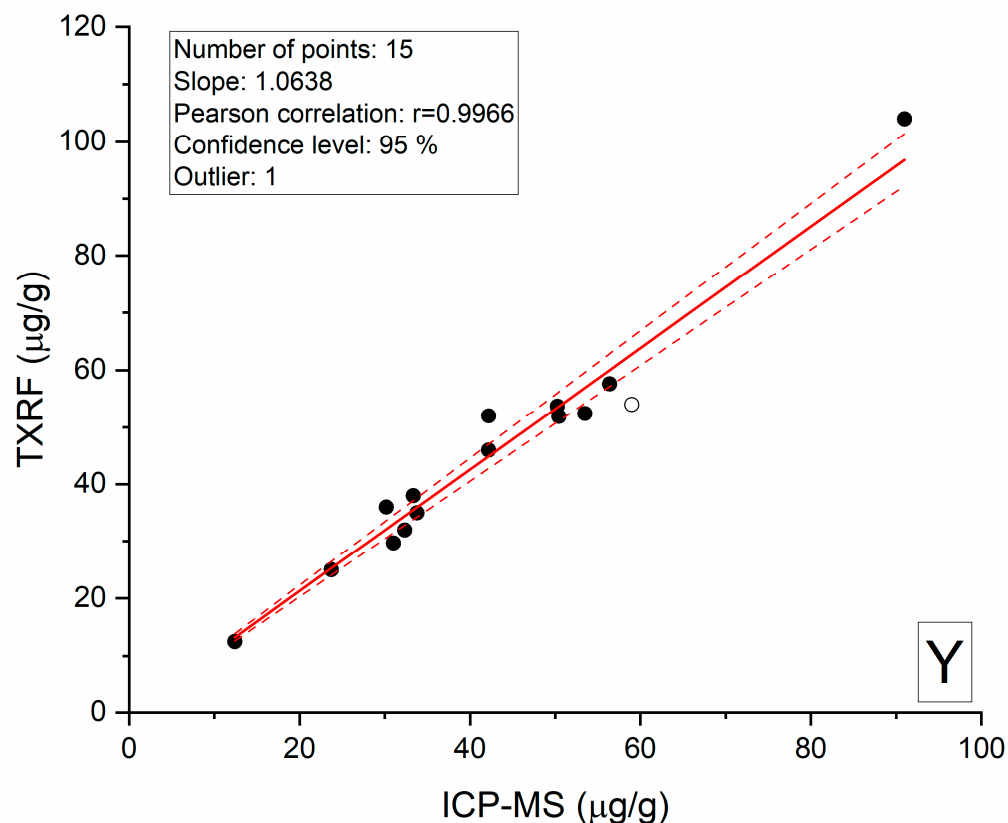


Figure 6. Comparison of the Y determination results in apatite samples by TXRF and ICP-MS. One outlier is highlighted by a blank circle. Dotted lines represent 95% confidence interval.

Figure 7 presents the Y/Ho—Sm/Yb ratio diagram for apatite from Slyudyanka, divided into three types. The values of apatite were compared to the reference values of terrigenous sediments, river water, modern seawater, and phosphorites, presented as clouds on the diagram. According to Zhang et al. [8], Y and Ho exhibit comparable vertical distribution patterns and ionic radii in the contemporary ocean; however, Ho is scavenged from the water column by particulates or ionic species almost twice as quickly as Y. The terrigenous protolith sediments of the Slyudyanka metamorphic rocks are characterized by relatively uniform, low Y/Ho ratios equal to 23–32 [37], and wide Sm/Yb ratios from 1 to 5, whereas modern seawater exhibits variable but generally higher Y/Ho ratios, commonly 50–100 [38], but narrow Sm/Yb ratios that do not exceed 0.97. Y/Ho ratios of river water are in the range 28–48, with almost no variation in Sm/Yb values (0.18–0.30) [14]. Phosphorites have the following characteristics: Y/Ho ratios of 43–75 and Sm/Yb ratios of 1–2 [39]. Figure 7 shows that the metamorphic apatite dots are inside the cloud of the phosphorites reference data. The Y/Ho ratio changes in seawater, which is why it accumulates in the phosphorites that form in seawater. Apatite of silicate–carbonate rocks has a chondritic Y/Ho ratio and resembles terrigenous rocks. REEs are immobile elements during high-grade metamorphism when little fluid is involved [40]. The remaining protholite Y/Ho signature in metamorphic apatite likely testifies that Slyudyanka high-grade metamorphism was dry, involving very little fluid. Yttrium and Ho usually fractionate in a hydrothermal environment [9]. Thus, it is surprising that hydrothermal–metasomatic apatite has Y/Ho ratios close to those of the terrigenous rocks. This example shows that the remaining chondritic Y/Ho ratio in apatite is not a universal fingerprint of hydrothermal origin. For apatite of silicate–carbonate rocks, there was a saline fluid that was different from seawater.

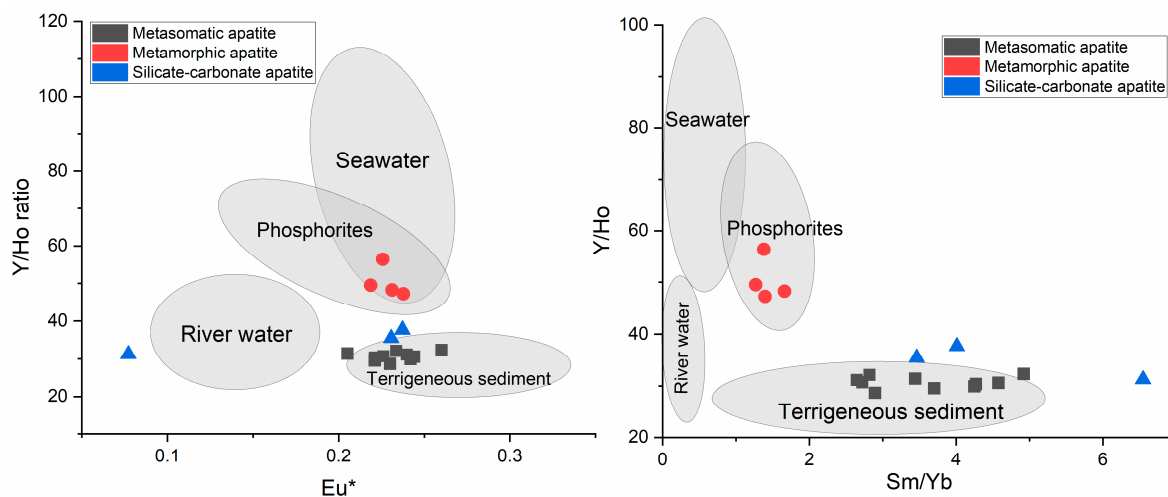


Figure 7. Y/Ho vs. Eu^* ($Eu/\sqrt{Sm \times Gd}$) and Y/Ho vs. Sm/Yb diagrams for different types of Slyudyanka apatite. The data were compared with the reference values for phosphorites [39], terrigenous protolith sediments of the Slyudyanka complex [37], modern river water [14], and modern seawater [41].

5. Conclusions

Three genetically distinct apatite groups from the Slyudyanka complex have been geochemically defined using TXRF and validated by ICP-MS, supported by multivariate statistical analysis (PCA, k-means). This classification provides a diagnostic fingerprinting framework for apatite in high-grade metamorphic terrains. (1) Hydrothermal–metasomatic apatite is identified by high Sr, Th, and As, and a chondritic Y/Ho ratio. (2) Massive metamorphic apatite from silicate–carbonate rocks is distinguished by extreme REE enrichment. (3) Metamorphic veinlet apatite from metaphosphorites is uniquely characterized by high U and Pb, low REEs, Sr, and Th, and a suprachondritic Y/Ho ratio. The key scientific advance is the demonstration that the suprachondritic Y/Ho signature—a hallmark of modern seawater—is preserved through granulite facies metamorphism. This finding is significant because it shows that the Y/Ho ratio in metamorphic apatite can act as a robust tracer of its sedimentary protolith. The anomaly is not “borrowed” but inherited; the metamorphic apatite retains the Y/Ho fractionation signature acquired by its marine sedimentary precursor (phosphorite) prior to metamorphism. Therefore, our work confirms the Y/Ho ratio’s utility as a powerful and resilient tool for reconstructing sedimentary provenance and paleoenvironments from detrital or metamorphic apatite.

The diagnostic geochemical framework established here has direct application in mineral exploration, enabling the efficient identification of apatite types with high REE potential. Furthermore, the validation of TXRF presents a practical, cost-effective analytical strategy for provenance studies and deposit characterization.

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Abbreviations

The following abbreviations are used in this manuscript:

TXRF	Total reflection X-ray fluorescence
ICP-MS	Inductively coupled plasma mass spectrometry
PCA	Principal component analysis
CA	Cluster analysis
REE	Rare earth element
CHARAC	CHARge-and-RAdius-Controlled
LA	Laser ablation
CAOB	Central Asian orogenic belt
ID-TIMS	Isotope dilution thermal ionization mass spectrometry
PTFE	Polytetrafluoroethylene
PAAS	Post-Archean Australian Shale
LOD	Limit of detection

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